How do you diagnose PANDAs?

Evidence searched on 2026-10-14 (PubMed, ClinicalTrials.gov, FDA drug labels, web)

Clinical summary

Bottom line Diagnosing PANDAS (Pediatric Autoimmune Neuropsychiatric Disorders Associated with Streptococcal Infections) currently relies on identifying **acute onset of obsessive-compulsive symptoms or tic disorders** alongside **evidence of streptococcal infection** (e.g., elevated ASOT titers, recent streptococcal throat infection) and **exclusion of other causes** (e.g., ASD, other tic disorders). Autoantibody screening (e.g., via nucleic acid programmable arrays) is an emerging but not yet standardized diagnostic tool, with 117 autoantibody targets identified in PANDAS patients distinct from ASD controls [1].

Most important recent finding Autoantibody screening identified **117 unique autoantibody targets** exclusively in PANDAS patients compared to ASD controls and healthy children, suggesting potential biomarkers for diagnosis [1].

Immediate context While clinical criteria (symptom onset, streptococcal link) form the diagnostic foundation, **no universally supported biomarkers or diagnostic algorithms** exist for PANDAS. Autoantibody testing remains investigational and not part of routine clinical practice.

Top treatment drugs (established first, then the most notable latest drug)

Drug	Indication named in the FDA label	Source
Dexamethasone	"Misce l l aneous: Diagnostic testing of adrenocortical hyperfunction, trichinosis with neurologic or myocardial involvement, tuberculous meningitis with subarachnoid block or impending block when used with appropriate antituberculous chemotherapy."	DailyMed

Latest findings

- **Autoantibody screening in pediatric autoimmune neuropsychiatric disorder associated with streptococcal infections (PANDAS)** (2026-04-29, **research-article**, *direct evidence*)

A study of **children with PANDAS compared to patients with autism spectrum disorder (ASD) and healthy controls** used unbiased autoantibody screening to identify 1,235 non-overlapping protein targets. **117 autoantibody targets were exclusively detected in PANDAS patients**, but profiles were highly heterogeneous, with no reliable diagnostic classifier due to cross-validation limitations. *[1]*

- **Immune dysregulation in pediatric tic disorders (including PANDAS)** (2026-03-10, **review-article**, *direct evidence*)

This review synthesized evidence on **neuroimaging, genetics (e.g., IL-1RN gene variations), and biomarkers** in pediatric tic disorders, including PANDAS. **No single diagnostic biomarker was established**, but immune-targeted therapies (e.g., IVIG, plasmapheresis) were highlighted as emerging approaches. *[2] (indirect evidence: different or related population)*

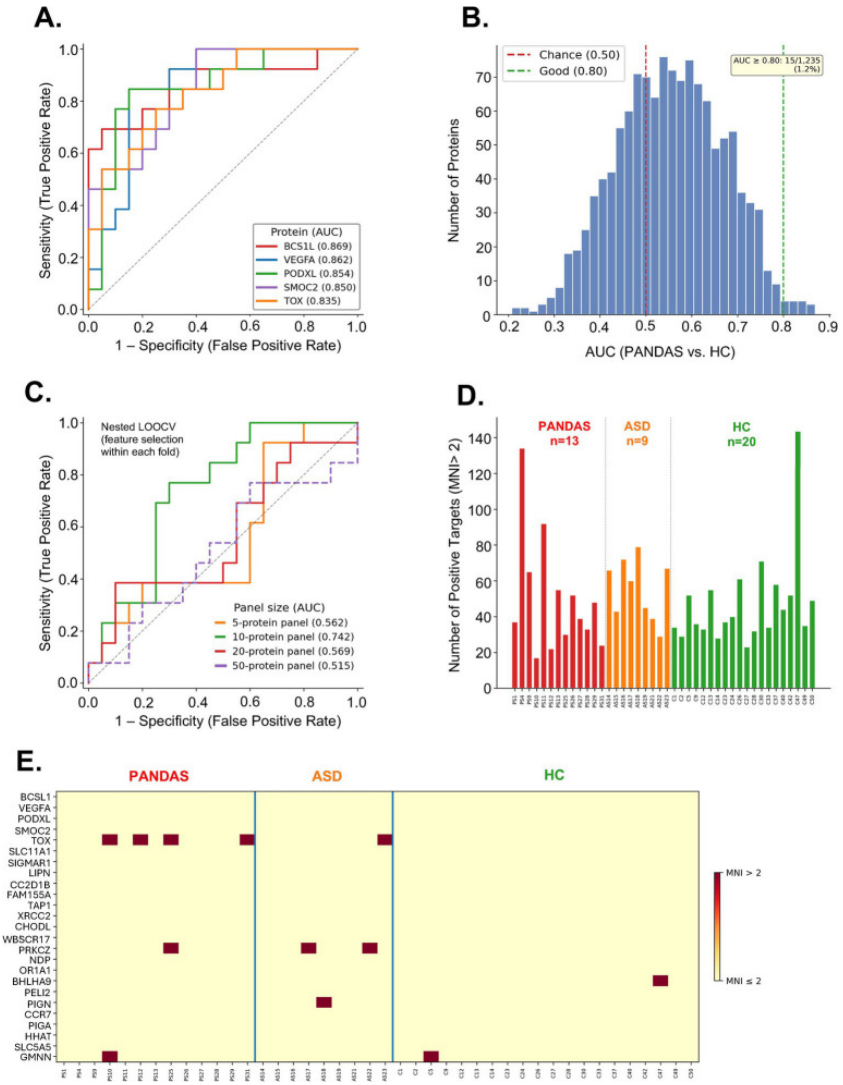
- **Pathways to PANDAS diagnosis in the UK** (2026-05-18, **research-article**, *direct evidence*)

Children aged 2–18 years with functional tic-like behaviors (FTLBs) were surveyed, revealing that **over half met PANDAS criteria** but faced delays, misdiagnosis (e.g., transient tics), and fragmented care. **No standardized diagnostic protocol was identified**, and private healthcare was accessed by 25% of diagnosed cases. *[3] (indirect evidence: different or related population)*

(No additional direct evidence retrieved for PANDAS diagnosis beyond 2026-04-29.)

Current evidence

Approach	Evidence level	What the evidence shows	Reference
Autoantibody screening using nucleic acid protein programmable arrays	**Emerging**	In children with PANDAS, 117 autoantibody targets were exclusively detected compared to ASD and healthy controls, but profiles were highly heterogeneous, with no reliable diagnostic discrimination achieved	[1]

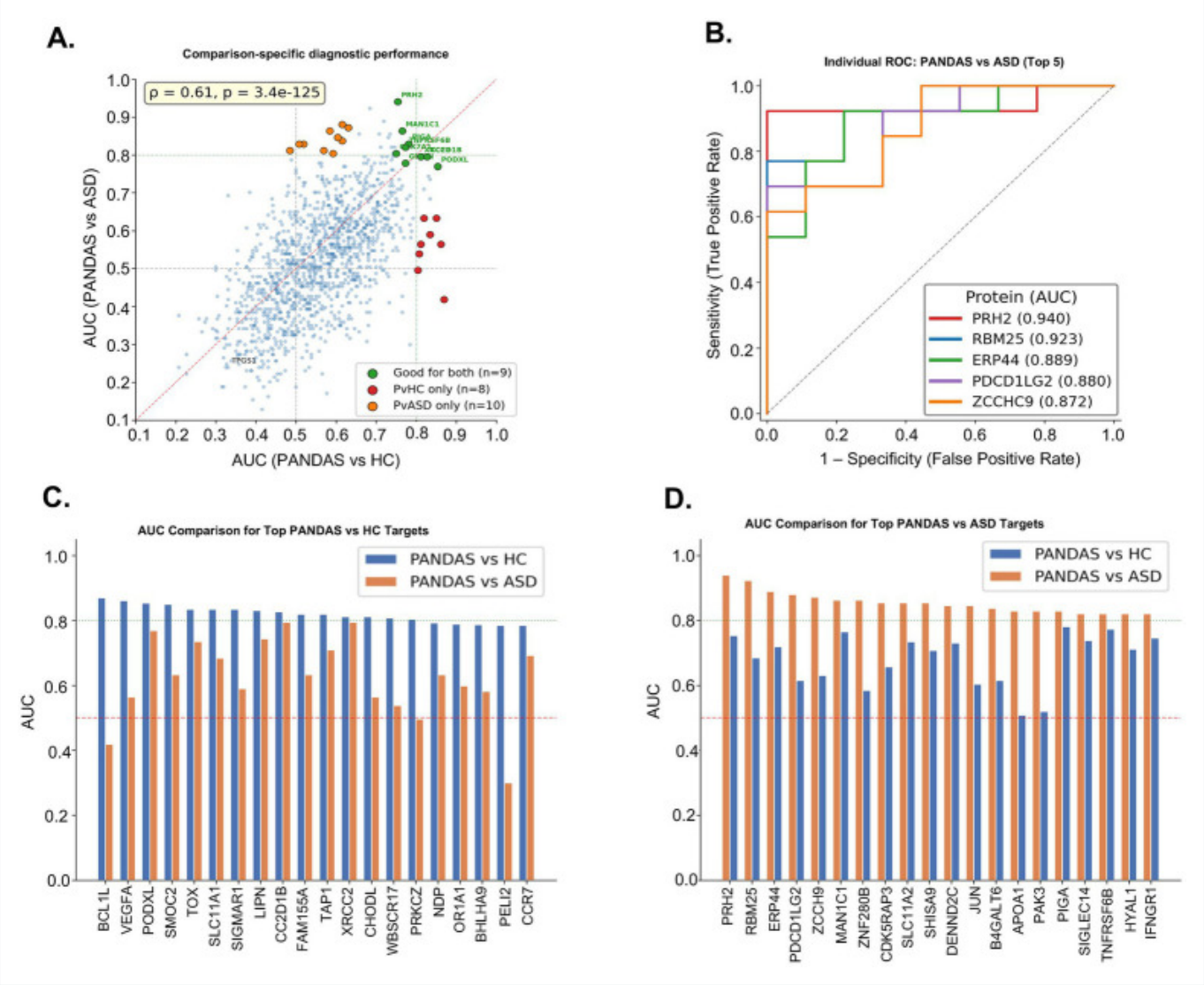


What this figure shows: A composite figure containing ROC curves for individual and panel protein targets, a histogram of AUC values, a bar chart of positive targets, and a heatmap of NAPPA autoantibody data.

How it helps with treatment options: This figure highlights that current autoantibody-based diagnostic approaches for PANDAS lack consistent accuracy, particularly when distinguishing PANDAS from other neuropsychiatric conditions like ASD.

Why this figure was chosen: The model's reason, from the caption: “ROC curves for the top five individual protein targets discriminating PANDAS from HC”. Checks passed: the paper was judged directly relevant to the question; the article states an open licence; the model's five-part relevance test was passed (helps answer, clinically meaningful, adds to the report, not decorative, shows the kind of information the question asks for); the picture is large enough to read on the page; a vision model confirmed that the image matches its caption and is readable.

Figure 4. Diagnostic performance analysis of NAPPA autoantibody data. (A) ROC curves for the top five individual protein targets discriminating PANDAS from HC, with AUC values in parentheses. The dashed diagonal line represents chance performance (AUC = 0.50). (B) Distribution of AUC values across all 1,235 proteins; red and green dashed lines indicate chance and good performance thresholds, respectively. Only 16 proteins (1.3%) achieved AUC >= 0.80. (C) ROC curves for multi-protein panel classifiers (5, 10, 20, and 50 features) using nested LOOCV with feature selection performed exclusively within each cross-validation fold. (D) Number of positive autoantibody targets (MNI > 2) per subject across PANDAS, ASD, and HC groups, demonstrating comparable autoantibody burden across groups. (E) Binary heatmap of the top 25 PANDAS-enriched proteins across all subjects, illustrating the sparse, patient-specific nature of autoantibody positivity.



What this figure shows: This figure compares protein-based diagnostic markers for PANDAS against healthy controls and ASD, showing that many proteins fail to reliably distinguish PANDAS from ASD due to overlapping reactivity. The scatter plot highlights that only a few proteins meet strict thresholds for discrimination, suggesting autoantibody testing alone may not reliably diagnose PANDAS.

How it helps with treatment options: This figure illustrates the diagnostic challenges in distinguishing PANDAS from ASD using autoantibody markers, emphasizing the need for alternative or complementary diagnostic approaches beyond protein-based tests.

Why this figure was chosen: The model's reason, from the caption: “Three-group diagnostic performance analysis. Scatter plot of individual protein AUC values”. Checks passed: the paper was judged directly relevant to the question; the article states an open licence; the model's five-part relevance test was passed (helps answer, clinically meaningful, adds to the report, not decorative, shows the kind of information the question asks for); the picture is large enough to read on the page; a vision model confirmed that the image matches its caption and is readable.

Figure 5. Three-group diagnostic performance analysis. (A) Scatter plot of individual protein AUC values for PANDAS versus HC (x -axis) versus PANDAS versus ASD (y -axis). Green points indicate dual discriminators (AUC >= 0.75 in both comparisons, n = 9); red points indicate HC-specific discriminators (AUC >= 0.80 vs. HC, <0.65 vs. ASD, n = 8); and orange points indicate ASD-specific discriminators (AUC >= 0.80 vs. ASD, <0.65 vs. HC, n = 10). Spearman correlation shown. (B) ROC curves for the top five individual targets discriminating PANDAS from ASD. (C) Paired AUC comparison for the top 20 PANDAS versus HC targets, showing their performance when tested against ASD (orange bars). Many targets exhibit substantial AUC drops, indicating shared reactivity between PANDAS and ASD. (D) Paired AUC comparison for the top 20 PANDAS versus ASD targets, showing their performance against HC.

Treatment Drug Landscape

Drugs used for treatment of this condition

Approved: an FDA label names this condition. Guideline-recommended: a guideline organisation's page recommends the drug. Used in practice: a retrieved review or page describes the drug for this condition, with no FDA label or guideline for it (see the category in Key evidence). Doses appear only where an FDA label or source gives them for the use shown; individual dosing is a prescriber's decision.

Drug	Indication	Dose / route (if sourced)	Key evidence
Dexamethasone	"Misce l l aneous: Diagnostic testing of adrenocortical hyperfunction, trichinosis with neurologic or myocardial involvement, tuberculous meningitis with subarachnoid block or impending block when used with appropriate antituberculous chemotherapy."	Route: Oral. No dosing subsection for this condition; the label's general dosing (not specific to this use): "DOSAGE AND ADMINISTRATION For Oral Administration The initial dosage varies from 0.75 to 9 mg a day depending on the disease being treated. It Should Be Emphasized That Dosage Requirements Are Variable And Must Be Individualized On The Basis Of The Disease Under Treatment And The Response Of The Patient. After a favorable response is noted, the proper maintenance dosage should be determined by decreasing the initial drug dosage in small decrements at appropriate time intervals until the lowest dosage that maintains an adequate clinical response is reached. (the label continues)"	FDA label. DailyMed label FDA-approved for the use shown; label effective 2025-10-29.

Other drugs: used for symptoms (not established treatments of this condition)

Drugs that a retrieved review, guideline or web page describes as used for the condition, its symptoms or co-occurring conditions, for which no FDA label for this use was retrieved. Drugs still being tested in trials are listed in the Clinical trials section below.

Drug	Indication	Dose / route (if sourced)	Key evidence
ibuprofen	beneficial for PANS/PANDAS patients (population: PANS/PANDAS patients) [associated symptom or comorbidity, not the condition itself]	Not stated in the retrieved source for this use	"OTC Anti-Inflammatory Drugs Some research shows that oral nonsteroidal anti-inflammatory drugs (NSAIDs) such as ibuprofen or naproxen may be beneficial for PANS/PANDAS patients." PANDAS Treatment Options
naproxen	beneficial for PANS/PANDAS patients (population: PANS/PANDAS patients) [associated symptom or comorbidity, not the condition itself]	Not stated in the retrieved source for this use	"OTC Anti-Inflammatory Drugs Some research shows that oral nonsteroidal anti-inflammatory drugs (NSAIDs) such as ibuprofen or naproxen may be beneficial for PANS/PANDAS patients." PANDAS Treatment Options

Recently approved or updated (FDA, EMA and CDSCO announcements)

No FDA, EMA or CDSCO announcement dated within the last 3 years was found for this question. This does not mean none exists.

Safety warnings (FDA boxed warnings)

Drug(s) (FDA label)	Boxed warning, in the label's words
naproxen	"RISK OF SERIOUS CARDIOVASCULAR AND GASTROINTESTINAL EVENTS Cardiovascular Thrombotic Events • Nonsteroidal anti-inflammatory drugs (NSAIDs) cause an increased risk of serious cardiovascular thrombotic events, including myocardial infarction and stroke, which can be fatal. This risk may occur early in treatment and may increase with duration of use [see Warnings and Precautions (5.1)]."

These are the labels' own boxed warnings, not a complete safety review: read the full label. No boxed warning on the retrieved label of: ibuprofen, Dexamethasone.

Clinical trials

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT06368726	tDCS or Transcranial Direct Current Stimulation; Risperidone	Phase 1/2	Completed (2026-09-10)	Completed; results not posted/retrieved.
NCT02676349	mFolfirinox; Chemoradiotherapy	Phase 2	Completed (2026-09-01)	Completed; results not posted/retrieved.
NCT04015596	Naproxen Sodium; Placebo	Phase 4	Terminated (2026-05-20)	Terminated early (Low recruitment); results posted (not retrieved here).
NCT04508530	Panzyga; Placebo	Phase 3	Completed (2026-01-07)	Completed; results posted (outcome not retrieved here). **Specific syndrome record; do not generalise to the whole condition.**
NCT01769027	Sertraline+Antibiotic (penicillin/azithromycin); Sertraline+placebo	Phase 2	Withdrawn (2023-02-21)	Registry status WITHDRAWN; no effectiveness conclusion.
NCT01281969	Gamunex Intravenous Immunoglobulin; Placebo	Phase 3	Completed (2020-03-17)	Completed; results posted (outcome not retrieved here).
NCT03348618	IVIG	Phase 4	Unknown (2019-11-19)	Registry status UNKNOWN; no effectiveness conclusion. **Specific syndrome record; do not generalise to the whole condition.**
NCT01617083	Azithromycin; Placebo	Phase 2	Completed (2018-04-10)	Completed; results posted (outcome not retrieved here).
NCT00001359	Penicillin or Placebo	Phase 2	Completed (2008-03-04)	Completed; results not posted/retrieved.

3 further registry record(s) were retrieved and are not shown (judged not to be treatment trials for this question).

Registry status shows that research is being conducted, not that a treatment works. Registry status is separate from published results.

Key studies

No retrieved record had all of: authors, year, study design, a stated result and a PMID, so none is listed.

Conflicting evidence

No direct disagreement was retrieved between studies on PANDAS diagnostic criteria. While **Eremija et al. (2026)** identified heterogeneous autoantibody profiles in PANDAS patients, **Ludlow et al. (2026)** reported clinical ambiguity in diagnostic labeling (e.g., PANDAS vs. transient tics) without conflicting biomarker findings. No studies directly compared diagnostic accuracy or supported a single gold-standard approach.

What remains under investigation

- **Diagnostic biomarkers**: No reliable autoantibody or biomarker panel has been supported by limited evidence for PANDAS, despite exploratory findings [1].
- **Neuroimaging correlates**: While basal ganglia changes are linked to tic disorders, their specificity for PANDAS remains unclear [2].
- **Clinical heterogeneity**: Subgroups (e.g., PANDAS vs. ASD overlap) require standardized diagnostic criteria to reduce mislabeling [3]. *(indirect evidence: different or related population)*
- **Longitudinal validation**: No prospective studies confirm whether early autoantibody detection predicts PANDAS progression or treatment response.
- **Immune profiling**: The role of *IL-1RN* gene variants and neuroinflammation in PANDAS diagnosis requires further mechanistic study [2]. *(indirect evidence: different or related population)*
- **Diagnostic algorithms**: No supported decision-support tools integrate clinical, immunological, or imaging data for PANDAS [1].

Evidence limitations

- Findings are drawn from abstracts, registry records and web pages, not full texts; study quality was not appraised from the full papers.
- Doses appear only where an FDA label gives them for this condition (section 2.1); for any other use, see a prescribing source. Approval dates appear only where a retrieved source states them.
- 7 retrieved record(s) were excluded because the relevance reading judged them not to answer the question.
- The web search timed out, so it may be incomplete; the absence of evidence cannot be concluded from it.
- The guideline-organisation search timed out, so it may be incomplete; the absence of evidence cannot be concluded from it.

Sources

1. Eremija J, Daines M, Rice S, et al. Unbiased autoantibody screening using nucleic acid protein programmable array in pediatric autoimmune neuropsychiatric disorder associated with streptococcal infections. *Frontiers in behavioral neuroscience*. 2026-04-29. PMID 42137259. DOI 10.3389/fnbeh.2026.1774848.
2. Sun X, Bai X. Immune dysregulation in pediatric tic disorders: mechanisms, biomarkers, and therapeutic frontiers. *Frontiers in immunology*. 2026-03-10. PMID 41939897. DOI 10.3389/fimmu.2026.1708940.
3. Ludlow AK, Anderson S, Robinson S, et al. Pathways to Care for Young People Diagnosed With Functional Tic Like Behaviours in the United Kingdom: An Online Mixed-Methods Survey. *Journal of paediatrics and child health*. 2026-05-18. PMID 42152473. DOI 10.1111/jpc.70437.

Other sources (regulators, guideline organisations, web pages, FDA labels)

- PANDAS Treatment Options
- FDA drug label: Dexamethasone (label effective 2025-10-29)

Research support, not a substitute for clinical judgement. All citations, trial IDs, links and numbers in this report were checked against the retrieved sources.

Figure credits

Figure 4: Unbiased autoantibody screening using nucleic acid protein programmable array in pediatric autoimmune neuropsychiatric disorder associated with streptococcal infections. research-article. PMID 42137259, DOI 10.3389/fnbeh.2026.1774848, PubMed Central (PMC13168177). Cited in this report as [1]. Licence: CC BY 4.0; original image, unaltered.

Figure 5: Unbiased autoantibody screening using nucleic acid protein programmable array in pediatric autoimmune neuropsychiatric disorder associated with streptococcal infections. research-article. PMID 42137259, DOI 10.3389/fnbeh.2026.1774848, PubMed Central (PMC13168177). Cited in this report as [1]. Licence: CC BY 4.0; original image, unaltered.

Each figure shows one study's result; it does not by itself establish that a treatment works or justify a recommendation.